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Landslides and debris flow triggered by the July 2024 extreme rainstorm in the Chooralmala watershed in Wayanad, India

Introduction

In the last week of July 2024, the Punchirimattom region in Wayanad experienced exceptionally heavy and continuous rainfall events, leading to two major landslides in the early hours of 30/07/2024 and a devastating debris flow that engulfed four places in the Meppadi Panchayath, namely, Punchirimattom, Mundakkai, Chooralmala, and Attamala. Over 200 buildings were washed away, and over 400 were damaged partially or fully, including two schools, one in Mundakkai and the other in Chooralmala. Three bridges were destroyed during the massive debris flow. This event has been described as one of the deadliest landslides of recent times. The loss of lives is currently reported as 251, and 47 individuals are still missing. Assessing the feasibility of continued habitation in the affected villages is crucial, given the significant debris deposited by the landslide.

The south Indian state of Kerala typically benefits from abundant rainfall during the monsoon season (June–September). However, recent climatic changes have led to a significant increase in rainfall accumulation in the region since August 2018 (Viswanadhapalli et al. 2019). What was once considered a blessing has become a concern as the frequency of landslides has risen dramatically. Data from the Geological Survey of India indicates that over 60% of rainfall-induced landslides in Kerala have occurred post-2015, with a particular concentration in the Wayanad and Idukki districts (Geological Survey of India 2024). One of the most catastrophic events occurred in Wayanad in 2024.

In response to the catastrophic disaster, the Kerala State Disaster Management Authority, Government of Kerala, formed a five-member expert committee to thoroughly evaluate the incident and recommend appropriate land use for the affected regions. This article's first author was a team member as a geotechnical expert. The field inspection was conducted in two phases during 3rd week of August 2024 and 1st week of September 2024, respectively. The investigation has provided critical insights into the causes and impact of the event. This article presents the details and analyses of the landslide using satellite data, Digital Elevation Models, and findings from the field investigations, offering a detailed account of the scale and severity of the disaster, along with the implications for future risk assessment and mitigation efforts in the region.

Study area

Wayanad, a prominent region along the Western Ghats in Kerala, is characterized by its diverse topography, with elevations ranging from approximately 37.5 to 2240 m above mean sea level. Figure 1 illustrates the terrain and administrative boundary map of Wayanad district, highlighting the location of recent debris flows. This figure underscores the elevated position of Vellarimala Hill compared to adjacent regions. The peak of Vellarimala (which lies on the southeast of the 2024 landslide crown) is 2240 m from MSL. The immediate hilltop (southern portion) above the crown has an elevation of 1863 m from MSL. Chooralmala is at an elevation of 853 m. The difference in elevation from the hilltop to plain land at Chooralmala is more than 1000 m within a short spatial distance. The elevation further comes down by approximately 400 m after the Soochippara waterfalls.

The landslide of 30/07/2024 occurred in the watershed of Puna Puzha, which originates from Vellarimala, having an elevation of 1863 m from MSL along the ridges. This area is prone to landslides, has high drainage density, and has vulnerable steep slopes. The Puna Puzha belongs to the Chooralmala watershed, encompassing an area of 18.30 sq. km. The region witnessed continuous rainfall in the last week of July 2024, with high-intensity rainfall on 29/07/2024 and 30/07/2024. A series of landslides occurred within a few hours in the early hours of 30/07/2024, leading to massive debris flow. The meandering course of the Puna Puzha River covering four locations of high inhabitation accelerated the impact of casualties. According to the Wayanad District Soil Conservation Office reports, the debris flow covered an area exceeding 85,000 sq m (Wayanad District Soil Conservation Office 2024). The debris flow originated at the coordinates 11°27′56.93″N, 76°8′9.85″E and extended over 7.8 km.

The rainfall data for this study was sourced from the 0.1° × 0.1° gridded dataset provided by the Indian Meteorological Department (IMD) (Rajeevan et al. 2008) and a local rain gauge station. Figure 2 depicts the spatial rainfall distribution during four significant landslide events in Wayanad district: the Kurchermala landslide on 09/08/2018, the Puthumala and Kavalappara landslides on 08/08/2019 (Jain et al. 2021), and the Mundakkai landslide on 30/07/2024. Notably, these landslides occurred on the mountain's eastern slopes, despite prevailing monsoon winds typically moving from the southwest to the northeast. The figure highlights the concentration of intense precipitation events in the affected areas, which triggered severe landslides and flooding. The rainfall patterns during these events were marked by prolonged episodes of heavy rainfall spanning over 3 days, resulting in significant cumulative precipitation. Figures 3 and 4 compare average rainfall data from the Kalladi station and the entire state for July 2024, providing insights into the high-intensity rainfall recorded on 30/07/2024. Figure 3b shows an accumulated rainfall of 2124.2 mm for July 2024 in the region. Antecedent rainfall effects were analyzed using two cases: a 2-day moving average and a 5-day moving average, both peaking on 30/07/2024 with values of 572.8 mm and 808.6 mm, respectively. The prolonged rainfall significantly increased pore

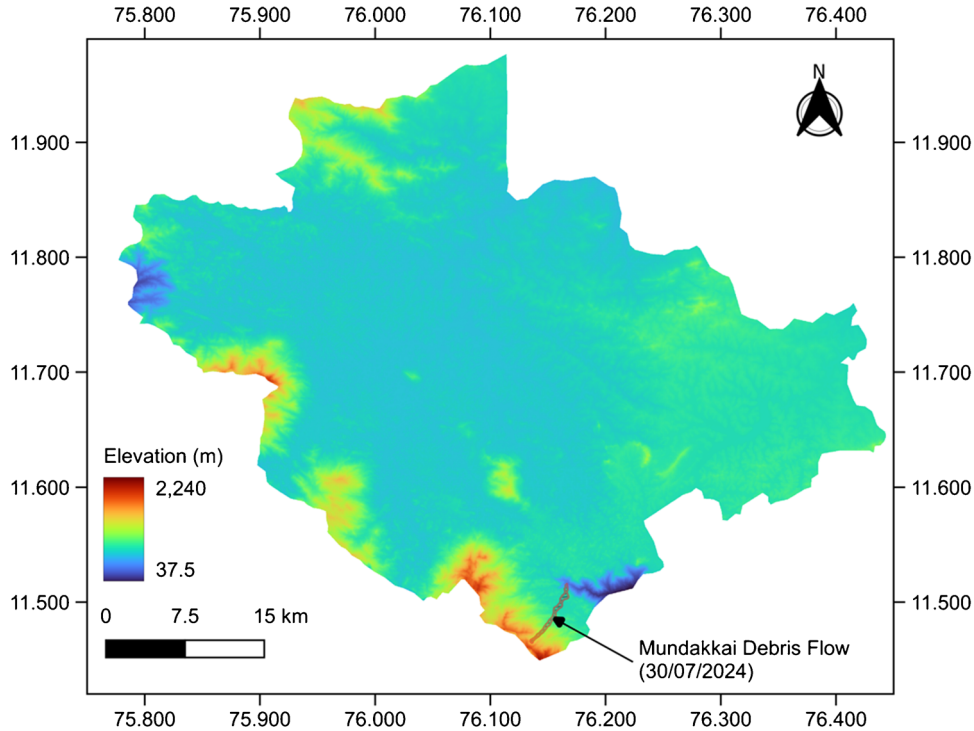

Fig. 1 Administrative boundary map of Wayanad with elevation

water pressure in the soil, reducing its shear strength and ultimately leading to catastrophic slope failures. These findings underscore Wayanad's vulnerability to landslides, driven by its topographic characteristics and the intensity of rainfall events.

The field investigation encompassed the region starting from Vellolippara, near the landslide's crown, and extended towards Punchirimuttom, Mundakkai, Chooralmala, and Attamala, covering an approximate distance of 7.2 km. Preliminary assessments indicated the absence of human activities or settlements in the area that could contribute anthropogenic factors to the landslide, identifying high-intensity rainfall as the sole triggering factor. The landslide originated in a forested area with no significant changes in land use in recent years. However, local residents reported historical alterations to the vegetation in the forest, which occurred when the land was acquired by the Government from Nilambur Kovilakom six decades ago. The hilltop is characterized by exposed rock outcrops, further defining the region's geomorphological setting.

A soil overburden of 6 to 8 m was observed at an elevation of 1200 m. Local accounts indicate that scar marks from a significant landslide in 2020, which originated in the same area, were extensively widened during the 2024 event. In the 2020 landslide, the bridge connecting Punchirimattom to the Vanarani estate was destroyed, and the debris flow extended to the Mundakkai region. Historical records also document a landslide in 1984 in the Aruna Puzha stream, which directly joins the Chaliyar River, within the Karimattam estate, located 3.7 km from the crown of the current landslide (Basak and Narasimhaprasad 1989; Geological Survey of India 2024). Additionally, a minor landslide occurred in 2018 within Estate No. 8, with debris flowing along Padavetti Puzha, which joins Puna Puzha at Chooralmala. Notably, the 2020 landslide was initiated at the same location as the 2024 event, creating scar marks along Puna Puzha that extended to Mundakkai. The 2024 event caused a significant widening of the stream, with debris flows traveling much farther downstream.

Geology and geotechnical parameters play a critical role in evaluating ground conditions in the study area. Soil samples were collected to analyze the geotechnical properties of the sandy silt predominant in the region, characterized by very low clay content. Figure 5a shows a soil sample taken from the exposed surface along the stream edge, identified as sandy silt with minimal clay content. Figure 5b illustrates samples collected from riverbed deposits, which consist primarily of silty sand with similarly low clay content. The sample collection locations, approximately 320 m from the nearest human settlement and extending toward the landslide crown, are marked in Fig. 5c, with the red-shaded area indicating the region impacted by the 2024 debris flow. Field observations highlight the dominance of softer rock formations, such as weathered and lateritized layers, which are highly prone to slope failures in the Wayanad region. Additionally, rocks exhibiting multiple brittle deformations or shearing further contribute to regional instability. The area is predominantly composed of foliated charnockite, often garnet-bearing (Fig. 5d). Figure 5e captures the first author beside a medium-sized boulder that detached from the hilltop during the landslide, causing significant damage to buildings in Mundakkai. These findings emphasize the critical interplay of geological and geotechnical factors in influencing slope stability in this area.

The loose sandy silt, characterized by low clay content, underwent liquefaction due to increased moisture content, soil saturation,

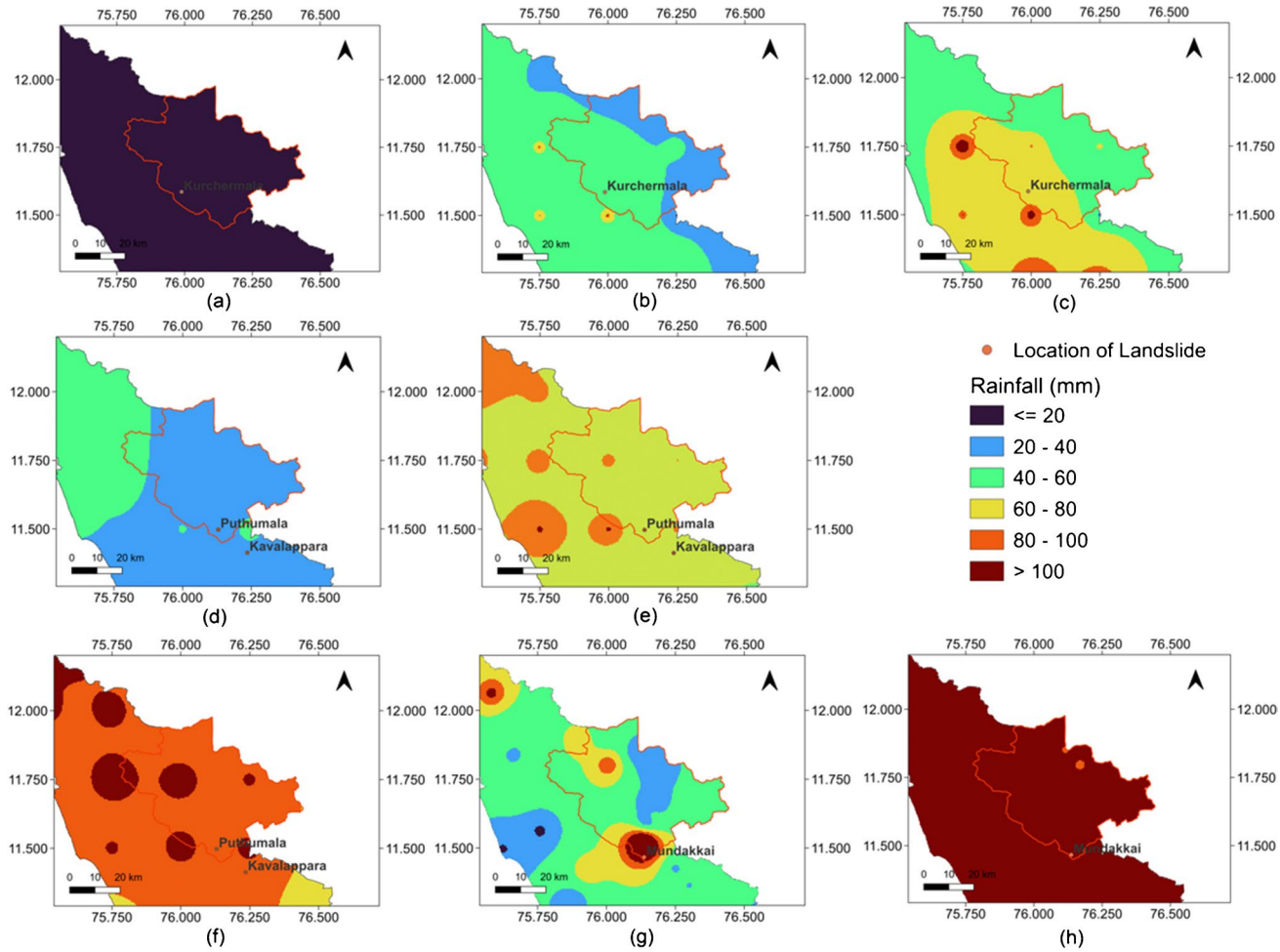

Fig. 2 Rainfall history of the locality corresponding to 2018 Kurchermala, 2019 Puthumala, 2019 Kavalappara, and 2024 Mundakkai, landslides for the dates **a** 07/08/2018, **b** 08/08/2018, **c** 09/08/2018, **d** 06/08/2019, **e** 07/08/2019, **f** 08/08/2019, **g** 29/07/2024, and **h** 30/07/2024

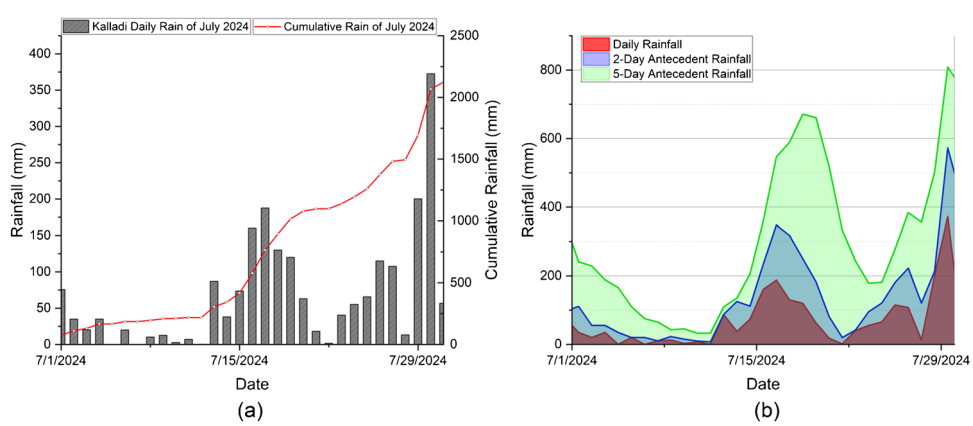

Fig. 3 **a** Rain gauge data of Kalladi station for July 2024. **b** Daily, 2 day and 5 day antecedent rainfall

and seepage forces induced by intense rainfall, ultimately triggering the debris flow. Embedded boulders within the soil were also dislodged and transported downstream during the event. Table 1 summarizes the index properties and other relevant parameters obtained from the collected soil samples. Figure 6 presents the particle size distribution of the soil, determined through dry sieve,

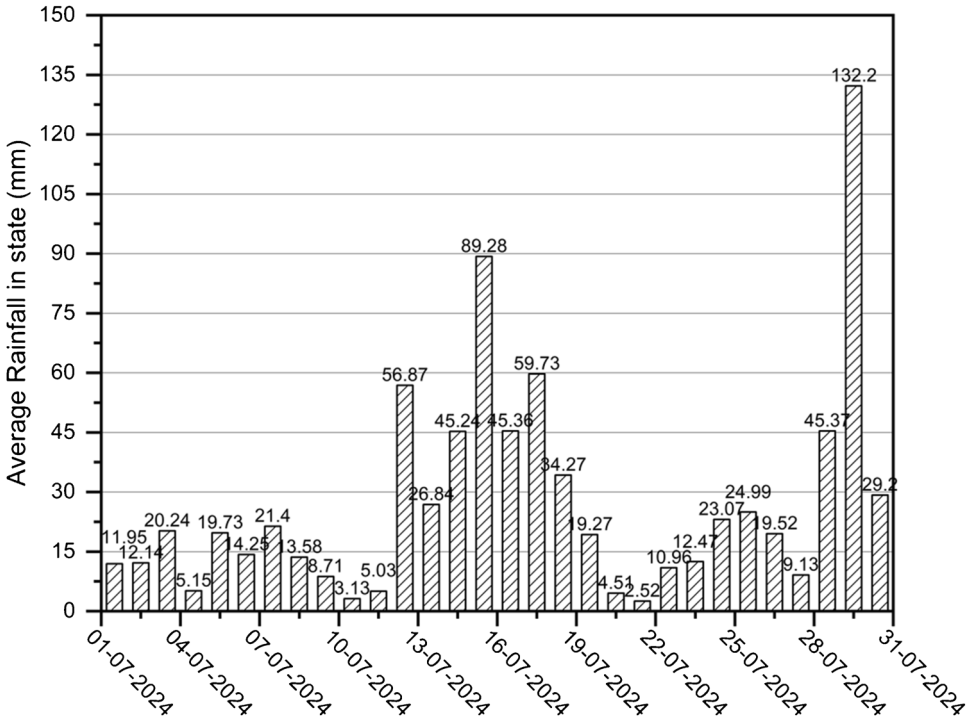

Fig. 4 Statistics of the state-wide average rainfall for July 2024

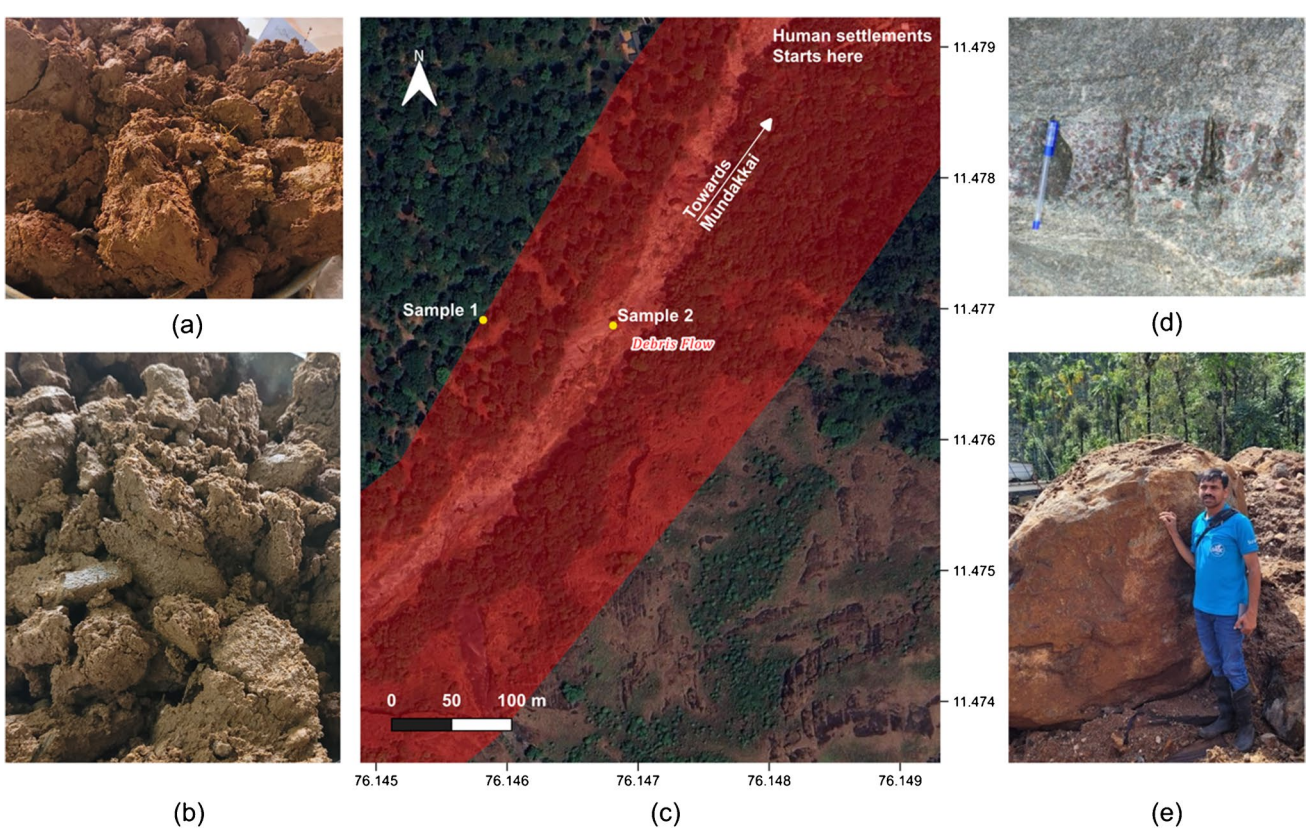

Fig. 5 **a** Soil sample from the stream edge (Sample 1). **b** Soil samples from the deposits in the streambed (Sample 2). **c** Location of the samples collected. **d** Foliated charnockite from site. **e** A medium-sized boulder from the site

Table 1 Basic geotechnical parameters of samples collected

Sample no	Liquid limit (%)	Plastic limit (%)	Field moisture content (%)	Specific gravity
1	46.9	34.3	32.47	2.63
2	26	22.5	19.91	2.62

wet sieve, and hydrometer analyses conducted in the geotechnical laboratory. The soil collected from the surviving stream banks was found to have a clay content of 29%, contributing to the stability of the thick soil overburden after the event. In contrast, the loose sandy silt sampled from the streambed exhibited a clay content of less than 10%, rendering it highly susceptible to liquefaction. These findings underscore the critical role of soil composition in influencing the stability and behavior of the terrain during extreme rainfall events.

Characteristics of debris flow and the societal impacts

The area encompassed by the slipped-out landmass is 86,000 sq m (NRSC 2024). The debris from the crown, which is a reactivation of an older landslide, has widened the river's course, causing a breach of its banks and damaging houses and other infrastructure along the banks. Figure 7 provides a detailed depiction of the debris flow's width, with estimates suggesting that the soil covering an area of 86,000 square meters was displaced during the landslide. The debris flow's reach extended to the Soochippara waterfalls, underscoring the event's immense scale and impact. Figure 8 captures the extensive destruction left after the debris flow, showcasing the severe damage to properties in the affected area. Figure 8a features an image taken near the Bailey Bridge, swiftly erected by the Indian Army to enable rescue operations and restore transportation in the disaster-stricken region. In this figure, the landside crown is also visible in the background. In Fig. 8b, a partially destroyed household is shown, illustrating the ongoing effects of the disaster, as the heavy rainfall persisted for more than a week after the debris flow had occurred. This continuous rainfall exacerbated the dire situation, highlighting the relentless nature of the natural forces.

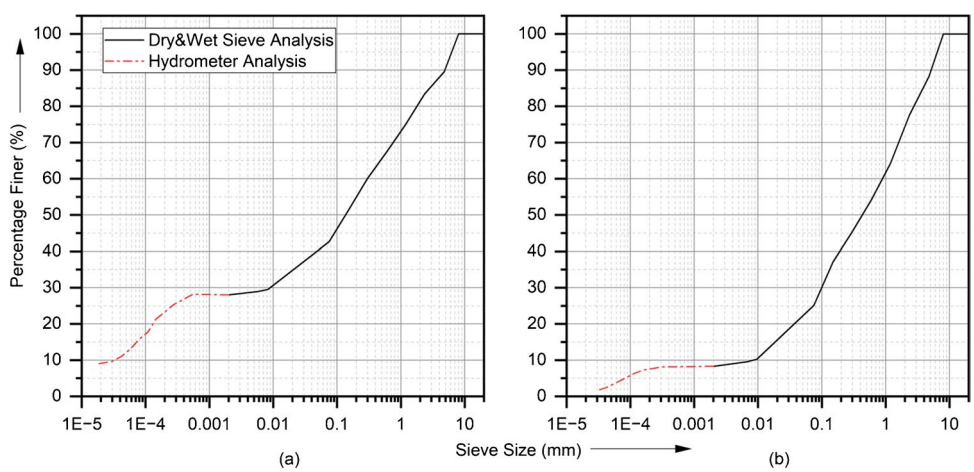

Fig. 6 Particle size distribution of **a** Sample 1 and **b** Sample 2

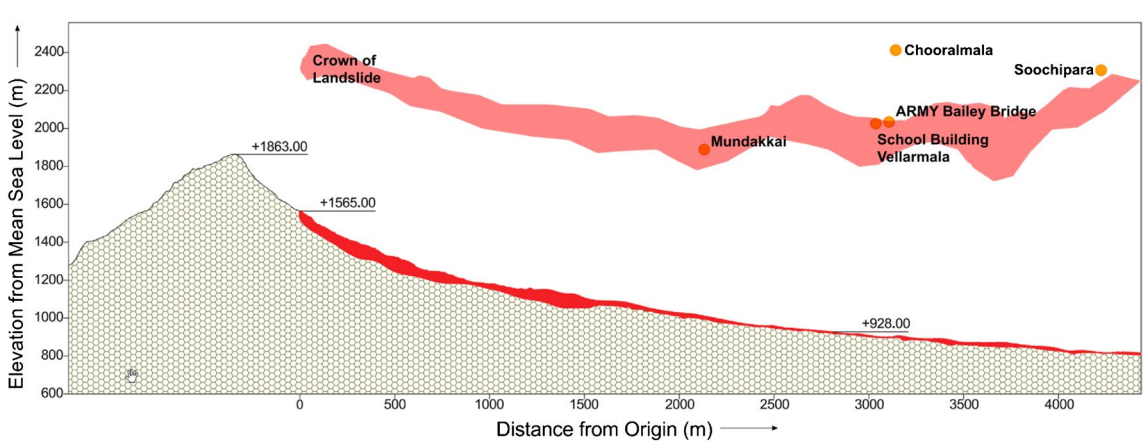

Fig. 7 Cross section of the landslide based on the digital elevation model data

Fig. 8 The pictures taken from various locations along the debris flow affected areas

Landslides are natural processes by which the earth seeks to regain stability, and typically, no further movement occurs once the soil debris has settled. However, a landslide transitions into a disaster when it intersects with human settlements or infrastructure, causing significant damage. The debris flow path in this particular event demonstrates that many buildings and infrastructure were severely impacted. Figure 8c offers visual proof of the liquefied state of the soil, confirming that the soil exhibited fluid-like behavior during the debris flow. This phenomenon is critical in understanding the landslide dynamics and its destructive potential. Figure 8d shows the view of the crown from Punchirimattom, and Fig. 8e shows the view of damage at Punchirimattom from where human settlement starts. In Fig. 8f, the first author is pictured standing next to a massive boulder that had rolled down the hill, a stark reminder of the serious threats posed by the debris flow. Although the boulder was not directly affected by the rainfall, the weakening of the supporting soil led to its eventual descent, contributing to the catastrophic events depicted in Fig. 8g, where the full scale of the disaster is evident. Figures 8h and i show government educational institutions roughly 200–300 m from the Puna Puzha tributary. Though seemingly at a safe distance, these structures were nonetheless severely damaged by the debris flow, rendering them unusable. This illustrates the far-reaching effects of the landslide, even on structures not directly in its path. Finally, Fig. 8j highlights the Bailey Bridge constructed by the Indian Army, which became a crucial element in the post-disaster recovery efforts. The bridge, erected within a single day, played an essential role in facilitating rescue operations and re-establishing vital transportation links in the aftermath of the disaster.

Local accounts indicate that the first significant sound was heard around 1:15 AM on 30/07/2024, likely caused by the planar failure of rock along a weak plane at the southern tip of the slope (elevation: 1565 m above mean sea level). A second major sound was reported at 1:40 AM, attributed to a soil slide on the southwestern tip. During the subsequent massive debris flow, the bridge connecting Punchirimattom-Mundakkai to the Vanarani estate—rebuilt after the 2020 event—was completely washed away. The variation in high-water marks observed in the Mundakkai region (elevation: 1200 m) suggests that debris, including boulders and wood logs, was deposited in the narrow valley of "Seethamma Kund." This blockage likely caused significant water accumulation, leading to overflow along the valley sides and extensive devastation eastward of the Mundakkai region. Eventually, the obstruction burst after 3:30 AM on 30/07/2024, under continuous heavy water flow, affecting downstream areas. The resulting debris flow widened the natural channels of the Puna Puzha tributary and transported large rocks and boulders downstream to Chooralmala, approximately 7.8 km away. The debris flow caused extensive damage to structures in Chooralmala town (elevation: 853 m) and continued to the Attamala region, ultimately reaching the Check Dam. These events highlight the magnitude and far-reaching impact of the landslide and associated debris flow.

The debris flow analysis was performed using the Rapid Mass Movement Simulation (RAMMS) software, a state-of-the-art tool designed specifically for modeling debris flows. This software leverages high-resolution Digital Elevation Models (DEMs) to provide accurate terrain representation. In this study, a 2.5-m resolution DEM obtained from the Bhoonidhi portal (NRSC 2024) was utilized and tailored to the study area to optimize computational efficiency while minimizing potential inaccuracies. The landslide crown was delineated and modeled using georeferenced drone imagery, ensuring precise integration with the DEM. Field reconnaissance established an average crown depth of 12 m. Friction parameters used in the simulation were adopted from prior research on the

Kurchermala landslide (Abraham et al. 2021), which occurred roughly 25 km from the current study area. The dry-Coulomb friction coefficient (μ) was set at 0.01, and the turbulent friction coefficient (ξ) at 100 m/s², based on the earlier study. The results of the analysis are illustrated in Fig. 9, while Table 2 provides a summary of key outcomes. The analysis estimates a soil release volume of 1,010,020.69 m³ and a maximum flow velocity of 30.20 m/s. According to the RAMMS output, the average debris flow height across the entire runout area is 7.29 m. However, specific regions near the crown exhibit flow heights exceeding 30 m. Field reconnaissance confirmed debris flow heights exceeding 20 m at certain locations near the crown, with a gradual reduction in height observed as the debris flow progressed toward the Chooralmala region.

Table 2 Results obtained from the debris flow analysis

Parameter	Value
Dry-Coulomb friction coefficient, μ	0.01
Turbulent friction coefficient, ξ (m/s²)	100 m/s²
Soil release volume (m³)	1,010,020.69
Overall maximum velocity (m/s)	30.2003
Overall maximum flow height (m)	62.7048
Overall maximum pressure (kPa)	1824.11

Conclusions

The 2024 Wayanad landslide, classified as a debris flow, was triggered by intense and prolonged rainfall, with over 500 mm of precipitation recorded within 48 h. This catastrophic event impacted a large area, spanning from Punchirimattom through Mundakkai to Chooralmala and Attamala, resulting in widespread devastation. To mitigate the risk of similar large-scale failures in the future, proactive measures, including the relocation of residents from high-risk zones to safer areas, are critical. Prevention is far more effective than attempting to resist or manage disasters of this scale. While this event was predominantly a natural hazard, the significant exposure of human settlements, particularly in floodplain regions, amplified its impact, leading to a catastrophic disaster. Further research is essential to better understand the unsaturated behavior of the region's soil, a factor that plays a key role in landslide susceptibility. Additionally, establishing localized weather stations to provide timely community alerts when rainfall exceeds threshold values is crucial for disaster preparedness. The unique ecological sensitivity and steep terrain of Wayanad necessitate micro-level assessments to create reliable landslide hazard maps with clearly demarcated runout zones.

This study highlights the following key findings and recommendations:

1. This article provided a comprehensive analysis of the 2024 Wayanad landslide and debris flow, employing satellite imagery, rainfall data, DEM, and detailed field investigations. Soil samples collected from the affected sites were subjected to laboratory tests, offering insights into the soil properties. A detailed timeline of events, alongside the extent of damage and destruction, has been documented.
2. A RAMMS analysis conducted using pre-determined friction parameters ($\mu = 0.01$ and $\xi = 100$ m/s²) closely replicated the

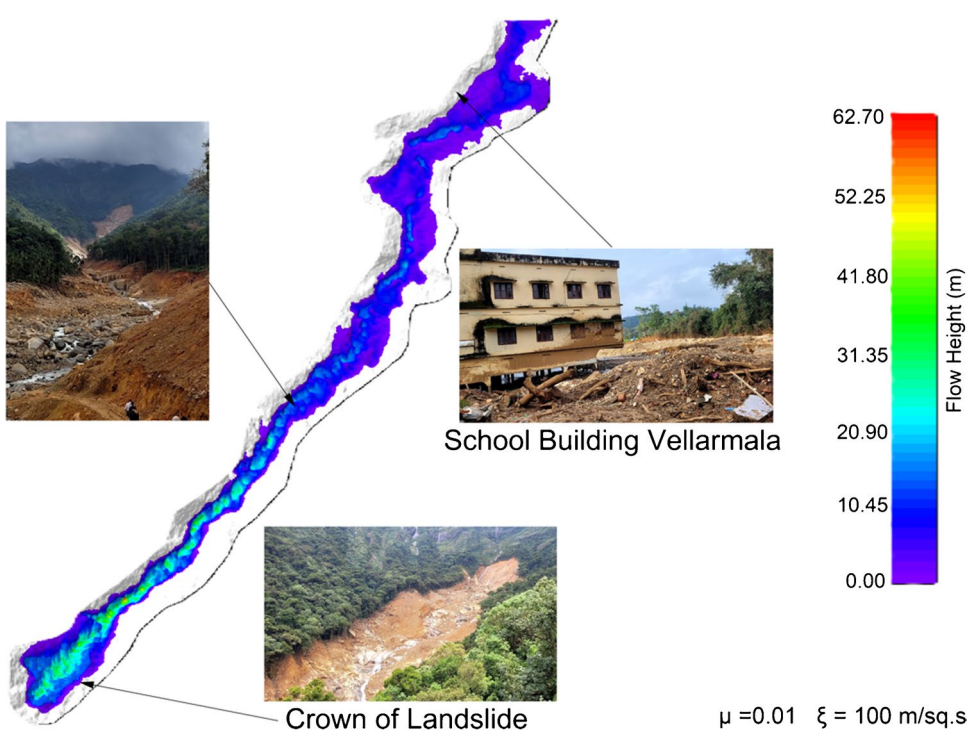

Fig. 9 Debris flow analysis results with $\mu = 0.01$ and $\xi = 100$ m/s.²

observed debris flow dynamics, providing valuable insights into the event's behavior.

3. Considering Wayanad's status as a landslide hotspot, micro-level risk assessments are imperative. Immediate relocation of residents from vulnerable areas must be prioritized before the next monsoon. High-risk zones should be declared unsuitable for residential development and, under strict regulation, may be repurposed for agricultural use where feasible.
4. Intense rainfall was identified as the primary trigger for the debris flow. Developing a rainfall threshold-based early warning system, informed by detailed investigations of similar events, is strongly recommended. Such systems would enhance community safety and contribute to sustainable regional development.

Implementing these recommendations will significantly reduce future risks and enhance resilience against natural hazards in Wayanad and similar ecologically sensitive regions. This study emphasizes the importance of integrating scientific insights with proactive disaster management strategies to protect both lives and livelihoods.

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Author contribution

Sreevalsa Kolathayar: site visit, collection of data, supervision, and editing of manuscript.
Varun Menon: debris flow analysis, preparation of manuscript, preparation of figures.
Priyajit Kundu: preparation of manuscript, preparation of figures.

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Data availability

The data underlying the findings of this study can be obtained from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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